Hamza Mumtaz

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EDUCATION & PROFESSIONAL DEVELOPMENT

- **Bachelor of Engineering** (Electrical Engineering) | Ontario Tech University (April 2026)
    - Minor: Sustainability and Energy Management
- **PLC Technician II Certificate** | George Brown College (August 2026)
    - Focus: Advanced PLC programming, ladder logic (IEC 61131-3), Rockwell Automation / Logix 5000, industrial automation networks, SCADA systems, HMI configuration, industrial troubleshooting, and electrical schematics.
- **Altium Essentials Certification** | Altium
    - Focus: Professional printed circuit board (PCB) design, PCBA, schematic capture, component management, and signal integrity.
- **Certification for Mechanical Design (CSWA)** | Dassault Systèmes
- WHMIS Certified | Workplace Health and Safety
- **Emergency First Aid & CPR Certified** | RedCross

TECHNICAL SKILLS

Control Systems & Automation: PLC Programming (Rockwell Automation/Logix 5000), HMI Development, Variable Frequency Drives (VFDs), Industrial Networks, Embedded Systems, PID Control, Closed-Loop Systems, Signal Conditioning, CAN Bus Architecture (SAE J1939).

Electrical & Hardware Engineering: Electronic Circuit Design, Schematic Capture, PCB Layout, Low-Voltage Automotive Wiring Harness Design, Power Distribution Networks, Component-Level Troubleshooting, Hardware Debugging.

Software & Engineering Tools: Altium Designer, SolidWorks, Siemens NX, MATLAB, Simulink, NI Multisim.

Fabrication & Hands-on Execution: Precision Wiring & Assembly, Soldering, Multimeters, Oscilloscopes, MIG/Stick Welding, Brazing, Precision Measuring (Calipers/Micrometers).

RELATED PROFESSIONAL EXPERIENCE

Systems Engineer/Race Systems Analyst, Mu Engineering, September 2024 - May 2026

- **Standalone Engine Management Systems:** Engineered PCBA with the use of **Altium Designer,** for a custom standalone engine management system utilizing an **ARM Cortex-M7** microcontroller, executing full lifecycle engineering requirements, schematic capture, multi-layer PCB layout, component-level PCBA assembly, and hardware bring-up of onboard **VR conditioners**, stepper drivers, and **signal-conditioning circuitry** to safely decode timing-critical crank/cam inputs with a ±0.5° tolerance up to 8,000 RPM.
- **Sensor & Timing Geometry Engineering:** Designed and manufactured a custom **hall-effect cam position sensor** and high-resolution reluctor wheel setup; calculated air-gap tolerances, **magnetic flux thresholds**, and tooth-angle profiles to eliminate signal noise during high-RPM transients.
- **Fluid Dynamics & Dry Sump Design:** Engineered a custom external **dry-sump oiling system**; modeled fluid scavenging efficiency, calculated pump displacement rates under continuous **high-G loading**, and machined custom distribution blocks to prevent oil starvation at high engine cornering loads.
- **Rotational Dynamics & Engine Tuning:** Executed a **balance shaft deletion** modification by calculating the removal of first and secondary torsional and reciprocating harmonics and parasitic rotational mass, effectively improving transient engine response while managing structural engine block harmonics.
- **Forced Induction Packaging & Electronics:** Modeled forced induction layouts within tight engine bay profiles using SolidWorks; designed and prototyped a digital electronic boost pressure gauge utilizing custom signal-conditioning circuitry and an analog-to-digital **(ADC) filtering algorithm**.
- **Suspension Dynamics & Simulation:** Conducted vehicle frequency simulations using **MATLAB**, cross-referencing motion ratios, damper valving profiles, and suspension compliance variables to achieve a precise target ride frequency of **1.7 Hz**.

Engineering Contractor, Mu Engineering, January 2021 - August 2024

- **Mechatronic System Prototyping:** Conceptualized and built an automated IoT mechatronic fluid/solids dispensing system; integrated analog load cells with microcontrollers and calibrated the system via standard mass measurements to ensure precise weight-based portion control.
- **Electromechanical CAD Design:** Formulated and modeled custom mechanical mounting brackets, gear profiles, and electromagnetic integration housings using SolidWorks, optimizing part geometry for strength, packaging constraints, and rapid 3D-printing/fabrication workflows.
- **Rapid Hardware Prototyping:** Designed and assembled numerous proof-of-concept electronics prototypes on perf-board platforms, validating circuit layouts, power distribution limits, and signal integrity constraints prior to final product definition.
- **Systems & Environmental Analysis:** Co-developed a data-driven highway traffic optimization framework for an engineering consultancy competition aimed at reducing regional GHG emissions across the Highway 401 corridor, cross-referencing vehicular throughput data against environmental impact models.

Industrial Electrical Assembler, EAM Mosca, December 2022 – May 2023

- **Industrial Control Panel & Enclosure Assembly**: Built and populated **industrial control enclosures, VFD/HMI panel boxes**, transformer enclosures, resistor boards, and junction boxes from engineering schematics; wired Phoenix Contact relays, terminal blocks, and custom pushbutton master control interfaces.

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- **Automation Hardware & Motor Integration:** Installed, routed, and wired **3-phase industrial motors**, high-voltage transformers, and complex sensor networks (including ultrasonic heads) for automated pallet-strapping machinery; integrated **Allen-Bradley PLCs**, Siemens components, and **ABB drives** into the machine chassis.
- **Containment & Power Routing:** Planned, bent, and ran **electrical conduit** and fished heavy-gauge power distribution and low-voltage signal cabling throughout heavy machinery frames, ensuring clean **signal separation** and structural strain relief.
- **Hardware Bring-up & Fault Diagnosis:** Executed point-to-point **continuity testing**, live circuit validation, and **multi-meter diagnostics**; isolated and resolved complex electromechanical machine faults, tracking down faulty wiring or hardware misconfigurations prior to final QA sign-off.

ADDITIONAL WORK EXPERIENCE

Automotive Mechanic/Operations & Safety Coordinator, Kenny U-Pull, May 2024 – Present
Customer Service Representative, Great Hobbies, April 2022 – July 2022
Quality Control & Inspection, General Motors, January 2022 – April 2022
Electronics Repair Specialist, Self-Employed, January 2014 - January 2021

SELECTED PROJECTS

Capstone Project – Autonomous Electric Vehicles - Electrical & Hardware Systems Integration Lead

- Distributed System Integration: Co-developed an autonomous control architecture using **ROS Noetic/Melodic** on **NVIDIA Jetson** edge platforms (**Orin NX, Nano**) interfacing with custom Atmel microcontrollers.
- Mixed-Signal Throttle Design: Re-engineered OEM analog throttles by designing **custom PCBs** and implementing a microcontroller-generated **high-frequency PWM** signal routed through a **low-pass RC filter** to recreate jitter-free ESC input voltages.
- Hardware Validation & Mapping: Utilized **oscilloscopes** to map the ESC's dynamic operating thresholds (**PWM 50–255**) for precise hardware-in-the-loop calibration.
- Rigid Mechanical Actuation: Designed and 3D-printed custom mounting structures in **glass-fiber reinforced PPS (PPS GF-20),** completely eliminating structural flex for high-torque **70 kg/cm steering** and **35 kg/cm braking servos**.
- Failsafe & Portability Testing: Implemented **hardware-level watchdogs** (0.25 throttle ceiling and **immediate LiDAR-bypassed emergency stops**) via a **6-field serial protocol**; successfully scaled and validated the system across two separate vehicle platforms.

Autonomous Control Surface Vehicle Prototype - Lead Embedded Systems & Hardware Designer

- Designed and implemented a **closed-loop control system** using an **ATmega328P** microcontroller and a **6-axis IMU (BMI160)** to regulate vehicle dynamics via real-time servo motor actuation.
- Resolved critical **I2C signal degradation** issues caused by parallel-wire electromagnetic interference and bus capacitance by redesigning wire routing and optimizing **hardware pull-up/divider networks**.
- Engineered a **dual-axis data acquisition** workaround to isolate vehicle yaw variance, successfully **bypassing I2C hardware address conflicts (0x68 mirroring)** to maintain stable data streams.
- Modeled system dynamics in **MATLAB** to simulate transient response and stability criteria, bridging the gap between theoretical control loops and physical hardware constraints.
- Diagnosed and calibrated power distribution anomalies, managing a **5.98V BEC line** to safely interface with **low-voltage 3.3V digital logic** components under dynamic load

Programmable Assistive Interface Device - Design & Project Management

- Co-engineered a 3x3 plug-and-play adaptive macro keyboard utilizing an **ATmega32U4** to provide custom computer input shortcuts for individuals with limited hand mobility.
- Implemented a **matrix layout** using **I/O multiplexing** to minimize microcontroller pin constraints; authored firmware to execute active key **debounce functions** to eliminate multi-click signal noise and maintain a 1:1 input response.
- Modeled the structural enclosure in **SOLIDWORKS** to seamlessly integrate the mechanical Cherry MX switch array, internal routing, and status LEDs; fabricated the physical chassis via **3D printing**.
- Managed project execution utilizing a structured engineering lifecycle, tracking milestones via **Gantt charts**, conducting TA design reviews, and validating system integrity through rigorous unit, integration, and **black-box acceptance testing**.

Custom Flight Control & Multirotor UAV Platforms - System Dynamics Project

- Designed, fabricated, and iterated four custom FPV multirotor platforms to analyze closed-loop stability, power distribution constraints, and real-world system dynamics under high-G conditions.
- Developed early prototypes utilizing **Atmel-based microcontrollers** interfaced with **MPU6050 IMUs** via I2C to execute localized orientation sensing, firmware-level PID control loops, and custom-soldered power distribution boards (PDB).
- Advanced to **ARM Cortex-F4 and F7 processing architectures**, utilizing **Blackbox telemetry logging** to analyze IMU/Accelerometer noise frequencies, tune dynamic PID loops (**P, I, D** coefficients), and configure electronic noise filtering to eliminate high-RPM mechanical resonance.
- Engineered a ruggedized, sub-250g platform reaching 114 km/h; optimized strict weight-to-power constraints and applied **silicon conformal coating**, validating control loop disturbance rejection and hardware insulation by maintaining stable flight through a severe thunderstorm.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- VP Ontario Tech Car Club (2023-2025)
- OTU Drone Design Lead (2019-2020)
- First Robotics Competition (2015-2018)